

Ex.No.10 Use Ghidra to disassemble and analyze the malware code.

Date:

Project Objectives

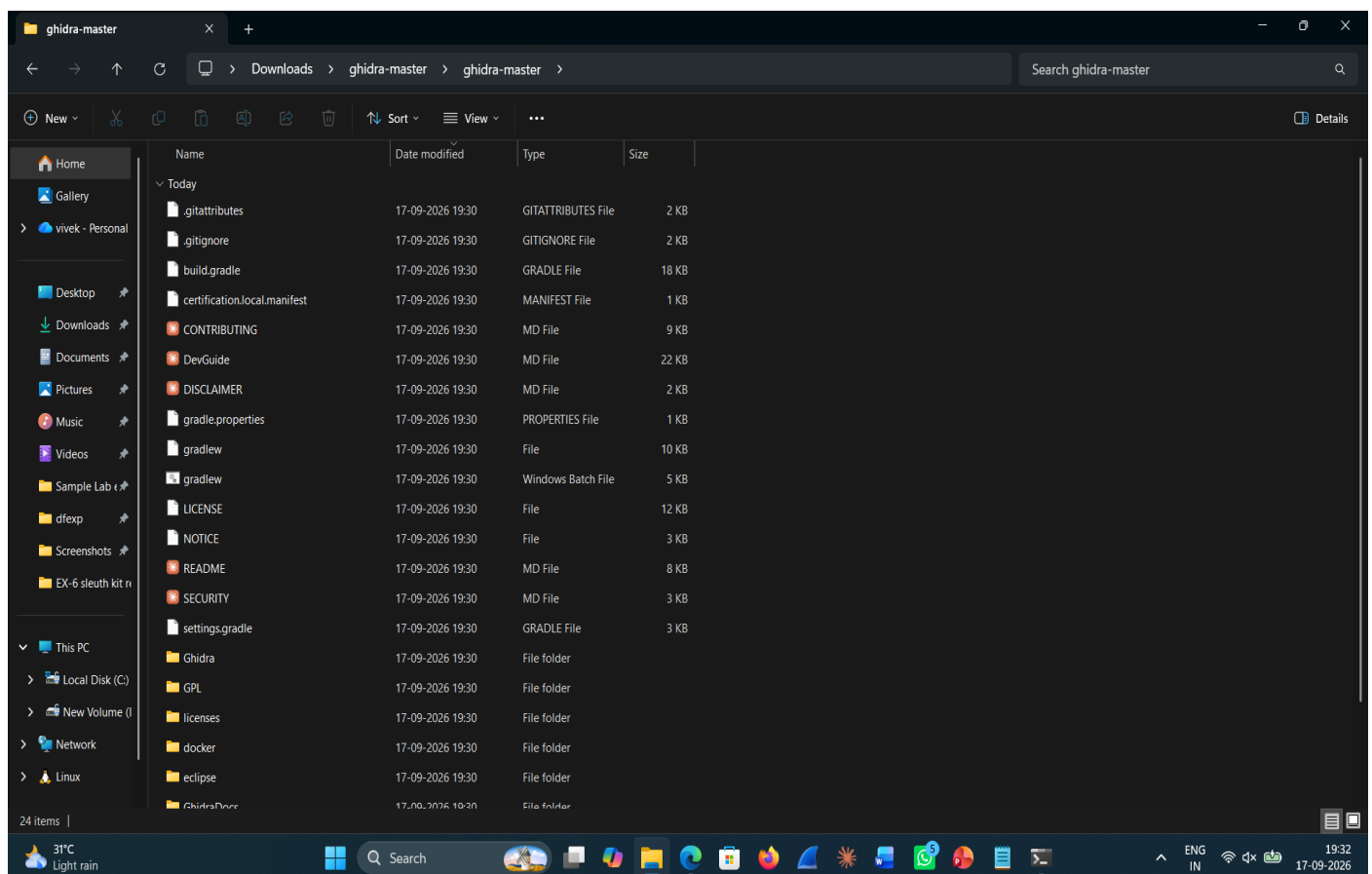
- To provide a hands-on guide to disassembling and analyzing malware with Ghidra.
- To enable readers to recognize common malware functionalities, such as persistence mechanisms, anti-analysis techniques, and network communications.
- To equip users with the skills to interpret low-level assembly code and understand high-level behaviours.

GitHub Repository Components

A GitHub repository for this project could include:

1. README.md

- Overview: Brief description of Ghidra, its installation, and how it is used for malware analysis.
- Requirements: List of requirements, including Ghidra, a VM setup, and any sample binaries (preferably benign samples or detailed steps to safely acquire controlled malware samples).
- Project Outline: A breakdown of the tutorial steps, linking to various sections or files within the repository.

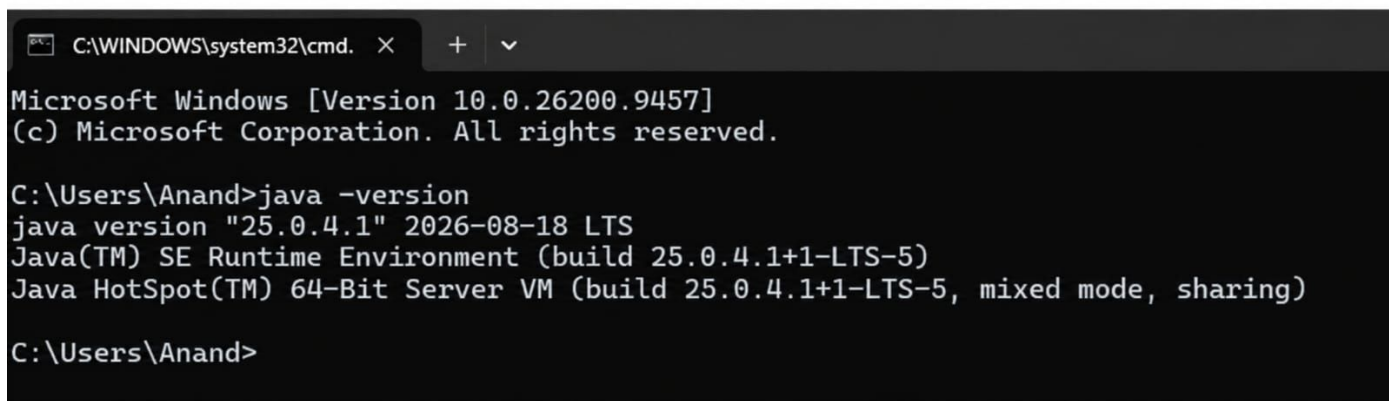


2. Tutorial Steps (Markdown or Jupyter Notebooks)

- Step-by-step instructions, broken down into digestible sections, such as:
 - Environment Setup: Instructions on setting up a virtualized and isolated environment.
 - Initial Analysis: How to load a binary into Ghidra, perform auto-analysis, and identify entry points.
 - Function Analysis: Methods for identifying and analyzing critical functions, decompiling code, and using Ghidra's cross-referencing features.
 - String and Import Analysis: How to locate and interpret strings and imports related to malicious behavior.
 - Advanced Techniques: Dynamic analysis, control flow graphs, custom scripting with Ghidra's Python or Java capabilities, and obfuscation detection.

3. Example Scripts (Python or Java)

- Scripts to automate repetitive analysis tasks, such as:
 - Extracting and listing all string references.
 - Identifying network-related functions or common indicators of malware.
 - Labeling functions and data segments based on known patterns (e.g., file system manipulation, registry edits).
- Script Examples: Python or Java code snippets demonstrating API usage in Ghidra (using Ghidra's Headless Analyzer, for example, for automation).



```
C:\WINDOWS\system32\cmd.  X  +  v
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Anand>java -version
java version "25.0.4.1" 2026-08-18 LTS
Java(TM) SE Runtime Environment (build 25.0.4.1+1-LTS-5)
Java HotSpot(TM) 64-Bit Server VM (build 25.0.4.1+1-LTS-5, mixed mode, sharing)

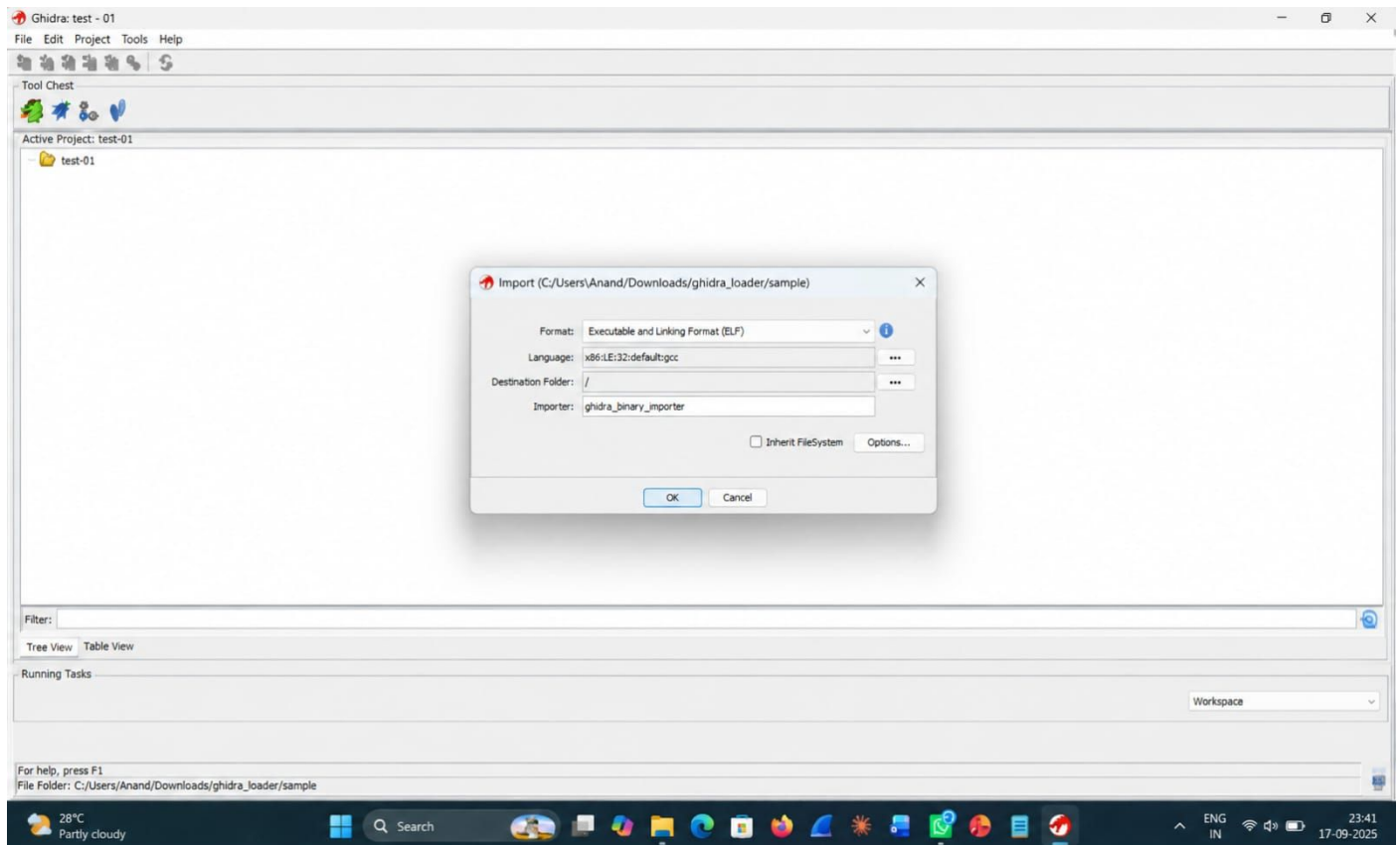
C:\Users\Anand>
```

4. Sample Data (Benign Binary or Hex Dump)

- Due to safety and ethical concerns, include only safe, benign binaries or a hex dump example rather than live malware. This can allow users to learn and practice without the risk of executing malicious code.
- Sample binaries designed to exhibit typical malware-like behaviors, such as:
 - Executing system calls.
 - Writing to files.
 - Sending network data.
- Alternatively, a dummy binary generator script that creates binaries with similar structures to malware but without harmful payloads.

5. Documentation and Comments

- Include explanations of the code within each script, detailing what each function does.
- Inline comments in disassembly views, providing readers with insights into assembly commands, control flow structures, and the decompiled code.



6. Report Template

- A reporting template for documenting findings, which could be in Markdown or PDF format. The template should cover:
 - Summary of Analysis: Brief description of the suspected malware.
 - Function Analysis: Summary of the key functions, their purpose, and the findings.
 - Behavioral Indicators: Summary of malware behaviors, such as persistence, file system manipulation, or network communications.
 - Recommendations: Suggestions on further actions, such as containment, eradication, or further analysis.

Example Repository Structure

plaintext

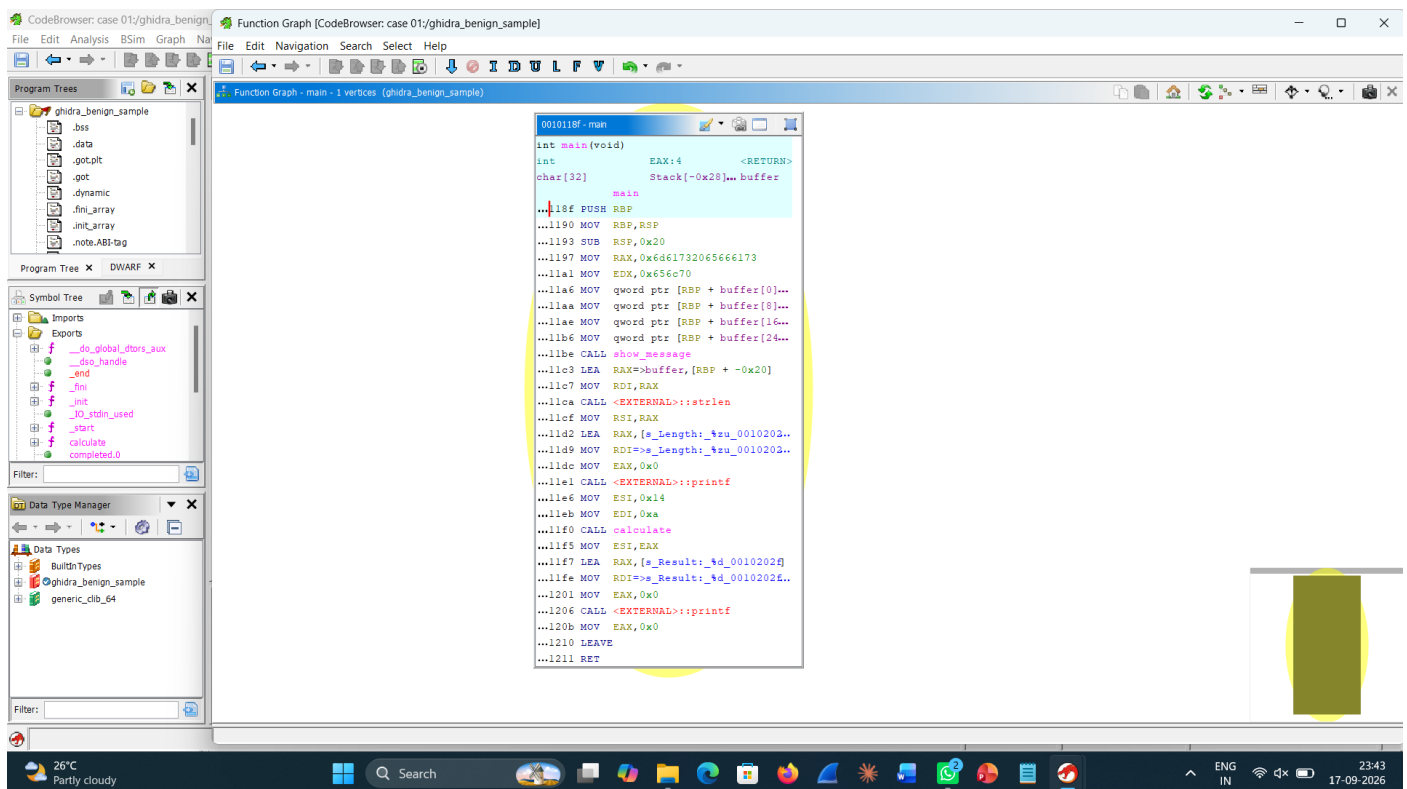
ghidra-malware-analysis-project/

|— README.md

|— docs/

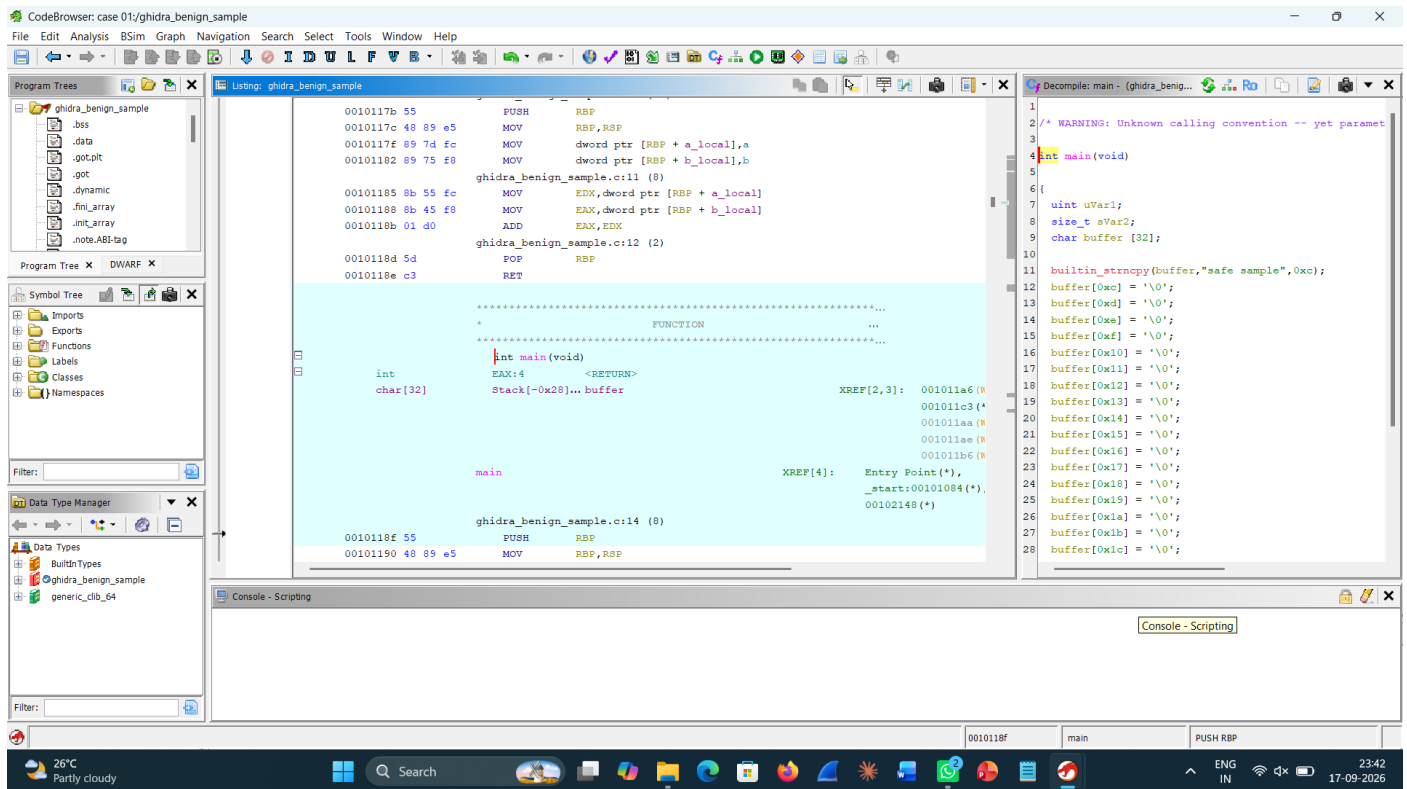
| |— Tutorial_Step1.md

- | | Tutorial_Step2.md
- | | Tutorial_StepN.md
- | scripts/
 - | | extract_strings.py
 - | | network_analysis.py
 - | | label_functions.py
- | samples/
 - | | benign_sample1.bin
 - | | benign_sample2.hex
- | templates/
 - | | analysis_report_template.md
- | screenshots/
 - | | function_graph_example.png
 - | | decompiler_example.png



Project Development and Future Work

- Expand Scripts: Continue adding Ghidra scripts to automate complex analysis tasks.
- Community Contributions: Encourage community members to contribute additional benign samples, analysis scripts, and documentation improvements.
- Advanced Topics: Include guides for advanced techniques, such as unpacking obfuscated code, identifying encryption routines, and using the P-Code emulator for runtime analysis.



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The binary file was successfully loaded and analyzed using Ghidra. Functions, strings, imports and decompiled code were examined to understand the program's behavior and identify possible malware-related characteristics.